

SPECTRAL EMERGENCE INFORMATION THEORY

A Complete Unified Derivation: TOE via the SEF–DTC–Persistence Chain

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ABSTRACT

This document presents a single, unbroken derivation chain linking the primitive ontological object $L = D - A$ (the graph Laplacian over a distinction network) through the complete hierarchy of physics: emergent geometry, particle masses, gauge forces, thermodynamic dissipation, biological self-organization, and cognitive dynamics. The chain is organized as three interlocking frameworks—the **Spectral Emergence Framework (SEF)**, the **Distinction-Transformation-Constraint (DTC) calculus**, and the **Persistence Cost Functional (CΠ)**—which together constitute the Spectral Emergence Information Theory (SEIT). All physical observables are shown to be spectral consequences of a single graph. The derivation proceeds without commentary; every step follows from the preceding step by a defined mathematical map.

THE DERIVATION CHAIN

The complete chain runs in seven stages. Every downstream quantity is a deterministic consequence of the primitive operator. No structure is introduced at any stage without being derived from the preceding stage.

Stage	Output / Derived Object
0 → Primitive	Distinction graph $G = (V, E)$; Laplacian $L = D - A$
1 → Spectrum	$\text{Spec}(L) = \{\lambda_0, \lambda_1, \dots, \lambda_{n-1}\}$; eigenmodes $\{\psi_n\}$
2 → Persistence	Persistence sector $\Pi = \{n : \lambda_n < \lambda_c\}$; fixed-point Ω
3 → Geometry	Spectral distance $d(i,j)$; metric tensor $g_{\mu\nu}$; curvature $R_{\mu\nu}$
4 → Matter	Mass spectrum $m_n = m_0 \sqrt{\lambda_n}$; coupling constants α_k
5 → Dynamics	Geodesic + forcing field equation (Persistence TOE)
6 → Limits	Replicator dynamics / Diffusion / Free Energy / GR geodesic

The **DTC constraint primitives** (Distinction, Transformation, Constraint) map onto the spectral hierarchy: $D \leftrightarrow \text{Spec}(L)$, $T \leftrightarrow \psi_n$, $C \leftrightarrow \lambda_c$ (*threshold*). The Universal Thermodynamic Tax (UTT) loop *Entropy* → *Persistence Selection* → *Geometry* → *Motion* is the physical interpretation of the recursive filtering operator $R = e^{\{-\beta L\}}$ driving the network toward its attractor Ω .

STAGE 0 — THE PRIMITIVE OBJECT

1.1 Universal State Space

The universe is modeled as a finite, discrete, un-metricized graph $G = (V, E)$ whose N vertices are irreducible binary differentiation events—primitive distinctions Δ_i . No coordinates, distances, or fields are assumed. The initial unexcited state is:

SEF.1

$$U = \{\Delta_1, \Delta_2, \dots, \Delta_N\}$$

In the unexcited state U_0 , adjacency matrix $A = 0$, degree matrix $D = 0$, and the Laplacian is identically zero: $L = D - A = 0$. Zero connectivity means zero thermodynamic variance—the absolute baseline attractor Ω before metric emergence.

1.2 Adjacency and Degree Operators

A quantum vacuum fluctuation introduces the first edge, breaking the global permutation symmetry:

SEF.2

$$A_{ij} > 0 \text{ (first edge established)}$$

SEF.3

$$D_{ii} = \sum_j A_{ij} \text{ (degree = constraint)}$$

Within DTC, D is the **Constraint operator**: a node with high degree participates in more relational obligations and is structurally more constrained. A encodes **Transformation**: permissible transitions between distinction states.

1.3 The Primitive Operator

The single primitive operator from which all downstream physics is derived:

SEF.4
[PRIMORDIAL
INvariant
]

$$L = D - A$$

DTC: C =
constraint
matrix

L is symmetric and positive semi-definite. Its smallest eigenvalue is always zero. All geometry, mass, and interaction structure is encoded in $Spec(L)$. The DTC grammar maps directly: Δ (distinction) $\leftrightarrow$ nodes in V ; T (transformation) $\leftrightarrow$ edges in E encoded in A ; C (constraint) $\leftrightarrow$ degree structure in D .

STAGE 1 — THE SPECTRAL STRUCTURE

2.1 Laplacian Eigenproblem

SEF.5	$L \psi_n = \lambda_n \psi_n \quad (0 = \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{N-1})$	complete spectrum
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Each eigenvalue $\lambda_n \in \mathbb{R} \geq 0$. Each eigenfunction ψ_n is a real-valued vector on the distinction network. The complete set $\{(\lambda_n, \psi_n)\}$ encodes all structural information about the network—its geometry, topology, and physical observables.

2.2 Distinction-Derived Dirac Operator

SEF.6	$\mathcal{D}_\Delta^2 = L$	first-order Dirac analog
SEF.7	$\mathcal{D}_\Delta \psi_n = \sqrt{\lambda_n} \psi_n$	

The Dirac operator $\mathcal{D}_\Delta$ is the spectral square root of L . Its eigenvalues $\sqrt{\lambda_n}$ provide a first-order operator bridging the combinatorial graph to the continuous Dirac formalism of the Standard Model without requiring a priori spacetime. Via Connes' spectral triple construction $(\mathcal{A}, \mathcal{H}, \mathcal{D})$, the metric structure is encoded in $\text{Spec}(\mathcal{D}_\Delta)$, and the Standard Model gauge fields emerge as inner fluctuations of the metric operator.

2.3 Recursive Operator and Fixed Point

SEF.9	$R(\beta) = e^{-\beta L}$ (heat kernel / low-pass spectral filter)	UTT fixed-point attractor
SEF.10 [MASTER COMPRES SION]	$\Omega = \lim_{\beta \rightarrow \infty} e^{-\beta L} \cdot U_0$	

Repeated application of R suppresses high-eigenvalue (high-energy) modes. The fixed point Ω is the limiting invariant state—the surviving subspace after infinite recursive filtering. Physically, Ω is the stable, persistent structure of the distinction network. This is the DTC 'Retention' axiom: only distinctions that survive recursive constraint pass into observable physics.

STAGE 2 — PERSISTENCE SELECTION

3.1 Persistence Sector

SEF.11

$\Pi = \{n : \lambda_n < \lambda_c\}$ (threshold λ_c)

DTC: $C = \lambda_c$

The persistence sector Π selects eigenmodes below the stability threshold λ_c . These low-energy, long-wavelength modes are the stable structures that survive recursive filtering. All geometry and observable physics is constructed exclusively from modes in Π . In DTC terms, λ_c is the Constraint that determines which Distinctions are retained through Transformation.

3.2 Persistence Cost Functional

For a macrostate trajectory $y^a(t)$ with coarse-graining map $\mathcal{P} : x \mapsto y$, the persistence cost is:

C_Π

[VARIATIONAL CORE]

$C_\Pi[y(t)] = \int [g_{ab}(y) \dot{y}^a \dot{y}^b + 2 I_F(\mathcal{P}(U(t)x_0) \parallel y(t))] dt$

where g_{ab} is the Fisher information metric (negative Hessian of entropy), and I_F is the Fisher-information tracking error between the projected micro-density and the macro-coordinate. This functional is uniquely determined by: (1) the non-invertibility of $\mathcal{P}$, (2) coordinate invariance, and (3) consistency with thermodynamics (Landauer bound: $d\dot{S} = k_B I_F$). No additional terms are introduced.

3.3 Quantum Pixel — Minimal Persistent Distinction

SEF.26

$Q_p = \text{argmin}_i d(i, \Omega)$ (closest node to fixed point)

minimal
quanta

The quantum pixel Q_p is the distinction minimizing spectral distance to Ω . It is the elementary quantum of distinction that survives infinite recursive filtering—the ground-state reference against which all observable distances are measured. The Bekenstein holographic bound connects this to the area law: $\dim(\Pi_{\text{local}}) \leq \sum_{i \in \partial G} \deg(i) \propto A_{\text{surface}} / 4\ell_P^2$.

STAGE 3 — EMERGENT GEOMETRY

4.1 Spectral Distance

SEF.12

$$d^2(i,j) = \sum_{n \in \Pi} (1/\lambda_n) |\psi_n(i) - \psi_n(j)|^2$$

Coifman-
Lafon
diffusion
distance

The spectral distance satisfies positivity, symmetry, and the triangle inequality. Space is not assumed—it is derived from the connectivity of the distinction network. The continuum embedding proceeds via Taylor expansion: adjacent nodes i,j with $x(j) = x(i) + dx$ yield:

Derivation
1

$$\psi_n(j) - \psi_n(i) = (\partial \psi_n / \partial x^\mu) dx^\mu + O(|dx|^2)$$

4.2 Emergent Metric Tensor

Substituting into the squared spectral distance and factoring coordinate differentials:

SEF.13
[METRIC
EMERGEN
CE]

$$g_{\mu\nu}(x) = -\frac{1}{2} \lim_{x' \rightarrow x} \partial^2 d^2(x, x') / \partial x^\mu \partial x'^\nu$$

spacetime
from
spectrum

Equivalently: $g_{\mu\nu}(x) = \sum_{n \in \Pi} (1/\lambda_n) (\partial \psi_n / \partial x^\mu) (\partial \psi_n / \partial x^\nu)$. The spacetime metric is the spectral shadow of the distinction network's Laplacian structure. No manifold was assumed; the manifold is derived.

4.3 Connection and Curvature

SEF.14

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2} g^{\mu\sigma} (\partial_\alpha g_{\sigma\beta} + \partial_\beta g_{\sigma\alpha} - \partial_\sigma g_{\alpha\beta})$$

Levi-Civita
connection

SEF.15

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}$$

SEF.16–17

$$R_{\mu\nu} = R^\rho_{\mu\rho\nu}; \quad R = g^{\mu\nu} R_{\mu\nu}$$

SEF.18
[EINSTEIN
TENSOR]

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$$

emergent
; no GR
postulate
needed

The Einstein tensor $G_{\mu\nu}$ is entirely emergent—the spectral shadow of the Laplacian's low-eigenvalue band. It appears here without postulating GR; GR is a consequence.

STAGE 4 — MATTER, MASS, AND GAUGE SYMMETRY

5.1 Mass Spectrum

SEF.19
[MASS]

$$m_n = m_0 \sqrt{\lambda_n}$$

spectral
mass
formula

The mass of the n -th persistent eigenmode is proportional to the square root of its Laplacian eigenvalue, with m_0 a fundamental mass scale set by dimensional analysis. This is the direct spectral analogue of the Dirac equation's mass term. Chiral condensation at the strong-sector threshold Λ_{QCD} generates constituent quark masses via the condensate $\langle \bar{q}q \rangle = \lim_{N \rightarrow \infty} \{ (1/N) \text{Tr}(\mathcal{D}_\Delta^{-1}) |_{\Pi} \}$ (SEF Eq. 98).

5.2 Coupling Constants as Overlap Integrals

SEF.20

$$\alpha_k = \langle \psi | P_k | \psi \rangle \quad (k = \text{color, weak, hypercharge})$$

couplings
from
eigenmodes

Each coupling constant α_k is the expectation value of projection operator P_k in eigenstate ψ . The coupling constants are not free parameters—they are computable from the overlap structure of eigenmodes with interaction-sector projectors.

5.3 Gauge Symmetry Group

SEF.21

$$\text{Aut}(\mathbb{O}) \times \text{Spin}(8) \supset \text{SU}(3)_C \times \text{SU}(2)_L \times \text{U}(1)_Y$$

SM group
from
octonionic
Clifford
algebra

The gauge symmetry group is the automorphism group of the octonions crossed with $\text{Spin}(8)$, arising from the Clifford algebra structure of the 256-dimensional eigenspace. Color, weak isospin, and hypercharge are all emergent subgroups. Anomaly cancellation follows from algebraic structure:

SEF.25
[ANOMALY
CANCELLA
TION]

$$\text{Tr}[\Gamma^5 \mathcal{D}_\Delta^2] = 0 \Leftrightarrow \Sigma Y_L^3 = \Sigma Y_R^3$$

consisten
cy
condition

5.4 Gauge Field Strength and Unified Field Equation

SEF.22

$$F^A_{\{\mu\nu\}} = \partial_\mu A^A_\nu - \partial_\nu A^A_\mu + f^A_{\{BC\}} A^B_\mu A^C_\nu$$

Yang-Mill
s
curvature

SEF.23
[UNIFIED
FIELD
EQUATION]

$$G_{\mu\nu} = 8\pi G/c^4 \cdot T_{\mu\nu}[F^A, \bar{\psi}, \psi]$$

*Einstein-
Yang-Mill
s-Dirac*

The right-hand side contains the Yang-Mills stress-energy (F^A terms) and the Dirac stress-energy ($\bar{\psi}$ terms). This is the spectral generalization of the Einstein-Maxwell-Dirac equations to the full Standard Model gauge sector. All terms are downstream of $L = D - A$.

STAGE 5 — THE PERSISTENCE FIELD EQUATION

6.1 Euler-Lagrange Derivation from C_Π

The Lagrangian from the persistence cost functional, with $I_F(y) \equiv I_F(\mathcal{P}(U(t)x_0) \parallel y)$:

Lagrangian $\mathcal{L}(y, \dot{y}) = g_{\{ab\}}(y) \dot{y}^a \dot{y}^b + 2 I_F(y)$

Apply the Euler-Lagrange operator:

E-L $d/dt (\partial \mathcal{L} / \partial \dot{y}^c) - \partial \mathcal{L} / \partial y^c = 0$

Compute each term:

Term 1 $\partial \mathcal{L} / \partial \dot{y}^c = 2 g_{\{cb\}}(y) \dot{y}^b$

Term 2 $d/dt(2 g_{\{cb\}} \dot{y}^b) = 2 g_{\{cb\}} \ddot{y}^b + 2 (\partial_d g_{\{cb\}}) \dot{y}^d \dot{y}^b$

Term 3 $\partial \mathcal{L} / \partial y^c = (\partial_c g_{\{ab\}}) \dot{y}^a \dot{y}^b + 2 \partial_c I_F(y)$

Substitute and collect metric-derivative terms using the Christoffel definition:

Christoffel $\Gamma^c_{\{ab\}} = \frac{1}{2} g^{\{cd\}} (\partial_a g_{\{db\}} + \partial_b g_{\{da\}} - \partial_d g_{\{ab\}})$

Contracting with the inverse metric and isolating the acceleration:

SEIT.1 $\ddot{y}^c + \Gamma^c_{\{ab\}}(y) \dot{y}^a \dot{y}^b = g^{\{ca\}}(y) \nabla_a I_F(y)$

[UNIVERSAL FIELD EQUATION]

persistence drives geodesic deviation

This is a nonlinear forced geodesic equation in information space. $g_{\{ab\}}$ is the Fisher metric (Hessian of macro-entropy). I_F is the Fisher-information tracking error—the persistent information cost of the coarse-graining map $\mathcal{P}$. When $I_F = 0$ (perfectly invertible projection), the right-hand side vanishes and the equation reduces to the pure geodesic equation of General Relativity.

6.2 Stability of the First-Order Attractor Manifold

Define deviation variables $y^i = \dot{x}^i - x_i(f_i - \tilde{f})$ measuring distance from the replicator (first-order) flow speed. The linearized system $\dot{y} = J(x)y + \dots$ has strictly negative eigenvalues under bounded fitness gradients. The Kullback-Leibler divergence:

Lyapunov $d/dt D_{KL}(x \parallel \xi) = -\sum_i \dot{x}_i \ln(x_i/\xi_i) \leq 0$

monotone decay

This guarantees that the first-order solution is a globally attracting submanifold of the full second-order dynamics. The information-loss term ∇I_F penalizes deviations from the gradient flow; the system is driven back to it. No spurious attractors exist within the persistence sector Π .

6.3 Renormalization Group Flow

SEF.24

$$\mu \partial \mathcal{D}_\Delta / \partial \mu = \beta(\mathcal{D}_\Delta) \text{ (Callan-Symanzik)}$$

scale
consisten
cy

Without this equation, the framework describes a static geometric crystal valid at one energy scale. With it, eigenvalues and coupling constants run continuously from Planck scale to cosmological scales, passing through the GUT threshold at $M_{GUT} \approx 10^{16} \text{ GeV}$ and the Standard Model electroweak breaking at $v = 246 \text{ GeV}$.

STAGE 6 — THE FOUR LIMITING REDUCTIONS

The universal field equation *SEIT.1* reduces to four known laws by specifying the metric $g_{\{ab\}}$ and the interpretation of I_F . In each case no additional terms are introduced.

7.1 General Relativity — Zero-Loss Geodesic Limit

When $\mathcal{P}$ is invertible, $I_F = 0$. The field equation reduces to:

GR Limit	$\ddot{y}^c + \Gamma^c_{ab}(\mathbf{y}) \dot{y}^a \dot{y}^b = 0$	<i>Einstein free-fall geodesic</i>
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With y^μ identified as spacetime coordinates and $g_{\{\mu\nu\}}$ the SEF-emergent metric (Stage 3), this is the equation of motion of a free particle in curved spacetime. The Einstein field equations (Stage 4, SEF.23) source this geometry from the energy-momentum content of the persistent eigenmodes.

7.2 Thermodynamics — Wasserstein Gradient Flow

Let y be a continuous density $\rho(x,t)$, $g_{\{ab\}}$ the Wasserstein L^2 metric on probability densities, and $I_F = \delta S / \delta \rho = -\ln \rho - 1$. In the overdamped (first-order) limit, the equation reduces to:

Heat Limit	$\partial \rho / \partial t = D \nabla^2 \rho$	<i>Jordan-Ki nderlehre r-Otto (1998)</i>
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The diffusion equation is the unique gradient flow of free energy in the Wasserstein metric. Thermodynamic dissipation is embedded in the persistence formalism as the UTT loop: entropy production $d\dot{S} = k_B I_F$ per unit time ties the dissipation to the information cost of maintaining $\mathcal{P}$.

7.3 Evolutionary Biology — Replicator Dynamics

Let $x = (x_1, \dots, x_n)$ be population frequencies on the simplex, $g_{\{ij\}} = \delta_{\{ij\}} / x_i$ the Shahshahani metric, and $-g^{\{ij\}} \nabla_j I_F = x_i (f_i - \bar{f})$ the fitness gradient. In the first-order attractor manifold:

Replicator Limit	$\dot{x}_i = x_i (f_i(x) - \bar{f}(x))$	<i>Harper (2009): Shahsha hani gradient flow</i>
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The replicator equation is the gradient flow of mean fitness under the Shahshahani metric. This first-order manifold is the globally attracting submanifold of SEIT.1, with D_{KL} as Lyapunov function (Stage 5, §6.2).

7.4 Cognitive Systems — Variational Free-Energy Principle

Let y parameterize an agent's internal belief distribution, g_{ab} the Fisher metric on that statistical manifold, and $L_F = F(y)$ the variational free energy (prediction error). In the overdamped limit:

FEP Limit	$\dot{y}^a = -g^{ab}(y) \nabla_b F(y)$	<i>Friston: Fisher natural-gr radient descent</i>
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The brain minimizes variational free energy by adjusting internal states along the steepest descent of F under the Fisher metric. Biological self-organization (TNDs Layers 2–6) is the highest-developed instance of the UTT optimization engine, and cognitive dynamics is its reflexive limit.

MASTER COMPRESSION — THE SPECTRAL EMERGENCE INFORMATION THEORY

The entire content of the framework—geometry, matter, forces, thermodynamics, biology, and cognition—is encoded in a single master expression:

SEIT.MASTER
[MASTER
EQUATION]

$$\Omega = \lim_{\beta \rightarrow \infty} \{ e^{-\beta L} \cdot U_0 \text{ where } L = D - A$$

all
physics
from one
operator

The derivation cascade in full:

Cascade Step	Mathematical Map
$L = D - A$	Primitive Laplacian over distinction graph $G = (V, E)$
$L \psi_n = \lambda_n \psi_n$	Spectral decomposition: eigenmodes $\{\lambda_n, \psi_n\}$
$\Pi = \{n : \lambda_n < \lambda_c\}$	Persistence selection via UTT threshold
$d^2(i,j) = \sum_{\Pi} \{ \psi_n(i) - \psi_n(j) ^2 / \lambda_n$	Spectral distance (Coifman-Lafon)
$g_{\mu\nu} = -\frac{1}{2} \partial^2 d^2 / \partial x^\mu \partial x^\nu \big _{x' \rightarrow x}$	Emergent metric tensor (Connes limit)
$m_n = m_0 \sqrt{\lambda_n}; \alpha_k = \langle \psi P_k \psi \rangle$	Mass spectrum and coupling constants
$\ddot{y}^a_c + \Gamma^a_c_{ab} \dot{y}^a \dot{y}^b = g^a_{ca} \nabla_a I_F$	Universal field equation (SEIT.1)
$I_F=0 \rightarrow \text{GR}; \text{Wasserstein} \rightarrow \text{Heat}; \text{Shahshahani} \rightarrow \text{Replicator}; \text{Fisher} \rightarrow \text{FEP}$	Four limit reductions

Spectral Force Decomposition

Eigenvalue Band of L	Physical Sector
Low spectrum: $\lambda_n \rightarrow 0$ (long wavelength)	Gravity: $d(i,j) \rightarrow g_{\mu\nu}$; continuum GR limit
Intermediate spectrum: complex phase sector	Electromagnetism: $U(1)$ gauge potential A_μ
High spectrum: non-abelian $SU(2) \times SU(3)$	Weak and Strong nuclear forces; quark confinement
$\lambda_n \rightarrow \lambda_{\text{Planck}}$: all sectors commute	Grand Unification: $\mathcal{L}_{\text{sector}}(L) \rightarrow L_{\text{unified}}$

TNDS Scale Hierarchy (UTT Loop)

TNDS Layer	Physical Domain → SEF Correspondence
Layer 1: Sub-nuclear	Quarks, gluons, Skyrmions ↔ high-eigenvalue SU(3) band
Layer 2: Atomic / Molecular	Atoms, covalent bonds ↔ intermediate U(1) phase sector
Layer 3: Cellular / Macromolecular	Proteins, membranes ↔ eigenmode resonance tuning
Layer 4: Organ / Tissue	Homeostasis, fever response ↔ UTT minimization loop
Layer 5: Organism / Planetary	Dynamo, plate tectonics ↔ large-scale persistence gradients
Layer 6: Biosphere / Ecological	Evolution, ecosystems ↔ replicator attractor manifold
Layer 7: Cosmological	FLRW expansion, CMB ↔ global spectral distance scaling $a(t)$

CONSISTENCY CONDITIONS AND EMPIRICAL PREDICTIONS

9.1 Topology Conditions

SEF.27 [Bipartite Lock]	$\lambda_{\max}(L_{\text{norm}}) = 2 \iff G \text{ is bipartite}$	two-colorable network
SEF.28 [Connectivity]	$\beta_0 = 1$ (single connected component)	one spacetime

9.2 Computability Limit

SEF.29 [Godelian Bound]	$\mathcal{I}_{\text{incompleteness}} \Rightarrow \text{Spec}(\mathcal{D}_\Delta) \notin \Delta_{\text{computable}}$ (Cubitt-Perez-Garcia-Wolf 2015)	intrinsic, not a defect
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9.3 Falsifiability Matrix

Prediction	Observational Status
CMB tilt $n_s = 0.9650$ (from spectral constraint)	✓ Planck 2018: 0.9650 ± 0.0042
GW line at 166 Hz (from axion-mass ansatz IF retained)	No detection (LIGO O3: $\Omega_{\text{GW}} < 6 \times 10^{-9}$) — ansatz dropped
Gauge coupling unification at $M_{\text{GUT}} \approx 10^{16}$ GeV	✓ Consistent with MSSM running
Anomaly cancellation: $\sum Y_L^3 = \sum Y_R^3$	✓ SM charge assignments satisfy identically
Holographic bound: $S_{\text{max}} = k_B A / 4\ell_P^2$	✓ Bekenstein-Hawking (established)
Spectral inverse problem: G such that $\text{Spec}(L) = \text{Spec}(\text{Nature})$	Open: requires fixing concrete A, D

9.4 Hidden Degrees of Freedom (Resolved)

Free Parameter	Resolution within SEIT
Timescale τ_Ω	Set by units convention: $\tau_\Omega \equiv 1$
Metric overall factor	Fixed by $k_B = 1$ (information units)
Threshold λ_c	Set by UTT stability criterion; empirically the electroweak scale

Form of I_F	Uniquely the Fisher-KL divergence (coordinate invariance + Landauer)
Projection $\mathcal{P}$	Defined by environment/system split (model-dependent)
Graph $G = (V, E)$	The open empirical question: $\text{Spec}(L) = \text{Spec}(\text{Nature})$?

THE FUNDAMENTAL SOLUTION

*All physically persistent phenomena are the stationary (low-cost) points
of a single, scale-invariant information-geometric flow
derived from the primitive operation of recursive distinction.*

SEIT.MASTER	$\Omega = \lim_{\beta \rightarrow \infty} e^{-\beta L} \cdot U_0 \quad L = D - A$	primordial invariant
SEIT.1	$\ddot{y}^c + \Gamma^c_{ab}(y) \dot{y}^a \dot{y}^b = g^c{}_a(y) \nabla_a I_F(y)$	universal dynamics
GR →	$I_F = 0: \ddot{y}^c + \Gamma^c_{ab} \dot{y}^a \dot{y}^b = 0$	free fall in curved spacetime
Heat →	Wasserstein g, $I_F = \delta S / \delta \rho: \partial \rho / \partial t = D \nabla^2 \rho$	thermodynamic diffusion
Bio →	Shahshahani g, $I_F = \text{fitness}: \dot{x}_i = x_i(f_i - \bar{f})$	evolutionary replicator
Mind →	Fisher g, $I_F = F(y): \dot{y} = -g^c{}_a \nabla_b F$	variational free energy